

Images in HTML

PW IOI • HTML Lecture 11–12 • Code in VS Editor → Output in Chrome

Images in HTML 1

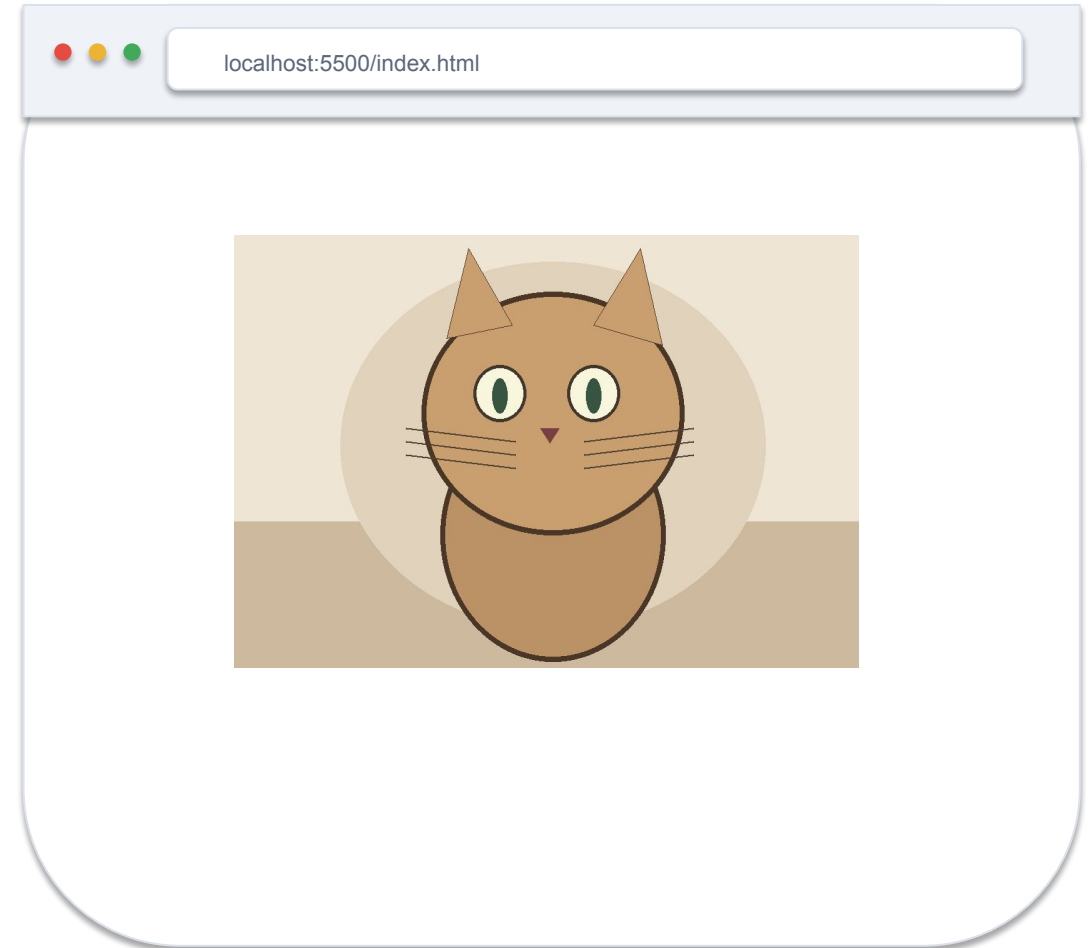
Formats • `<img>` • `src` • `width` • `height` • `alt` • `title` • legacy attributes

Images in HTML 2

Images in `<main>` • `crossorigin` • `decoding` • `fetch priority` • `srcset` • image cards

Images in HTML 3

Amazon-style homepage • fluid images • background-image cards



Types of Images and Their Use Cases

Choose the format based on the type of visual and the job it needs to do.

JPG / JPEG

Photos

Good compression for photographs. No transparency.

PNG

Logos / screenshots

Sharp graphics and transparency support.

GIF

Simple animation

Can animate, but limited color range.

SVG

Icons / logos

Vector format. Remains sharp when resized.

WebP

Modern web images

Good quality with smaller files in many cases.

AVIF

Highly compressed images

Modern format with strong compression and good quality.

 Tag

The img element embeds an image inside the webpage.

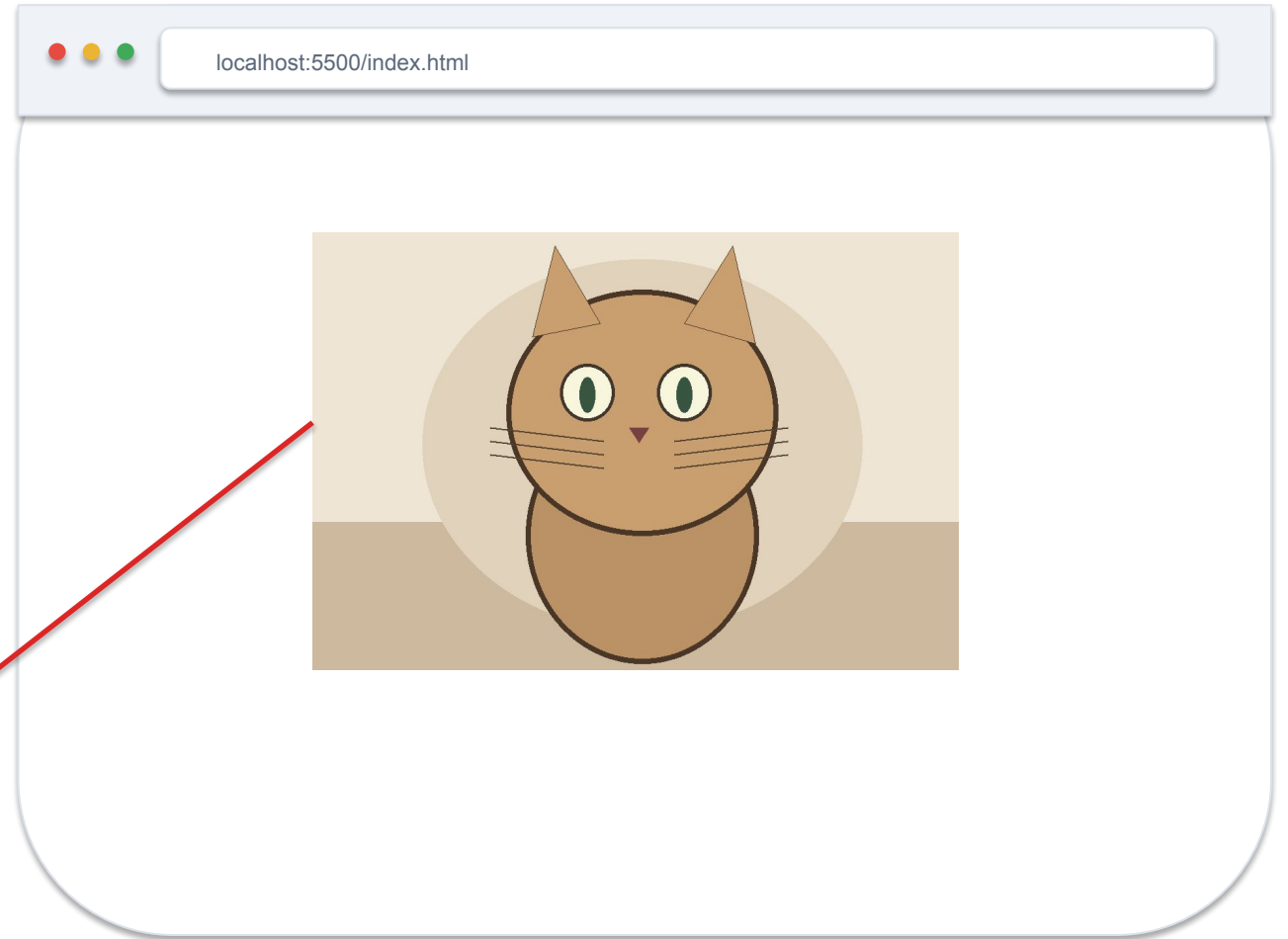
Definition

The tag tells the browser to load an image file and display it at that position in the document. It is a void element, so it does not need a closing tag.

```
index.html  
  

```

 creates this image element



src Attribute

src = source: the location of the image file.

The browser reads src, finds that file or URL, downloads it, and displays it. A wrong path means the image cannot be loaded.

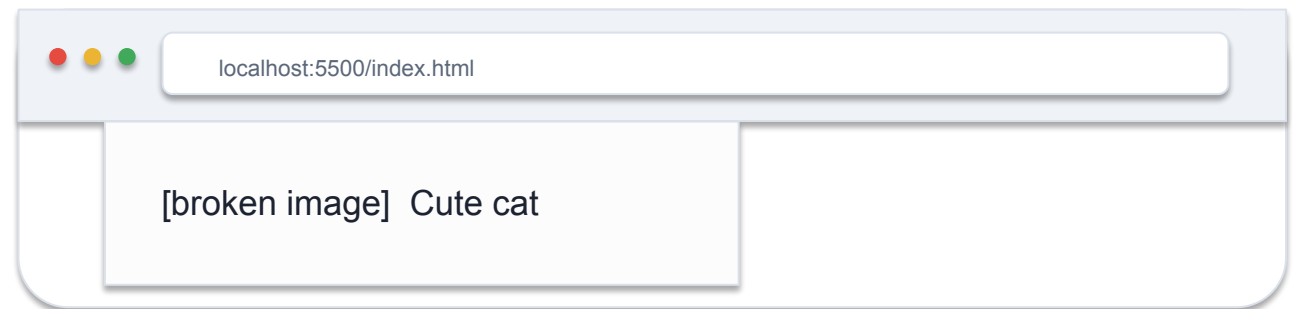
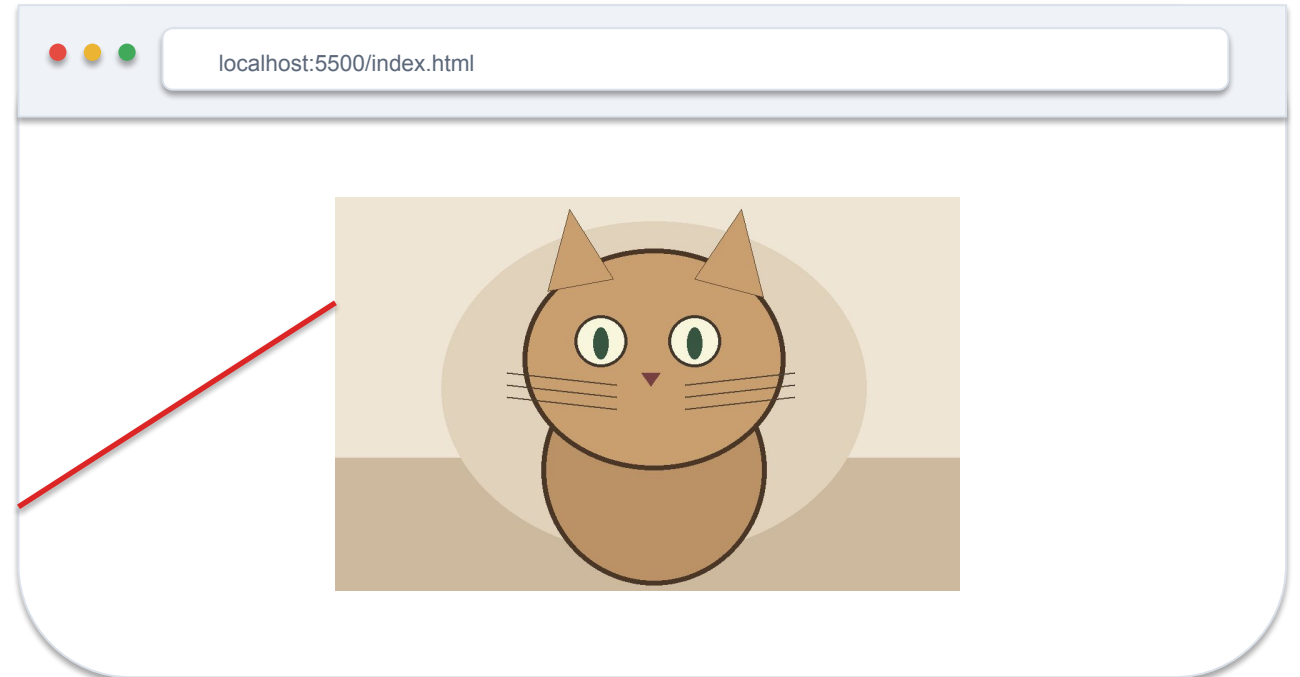
```
... index.html  
  

```

src="cat.jpg" → loads this file

```
... index.html  
  

```



width and height Attributes

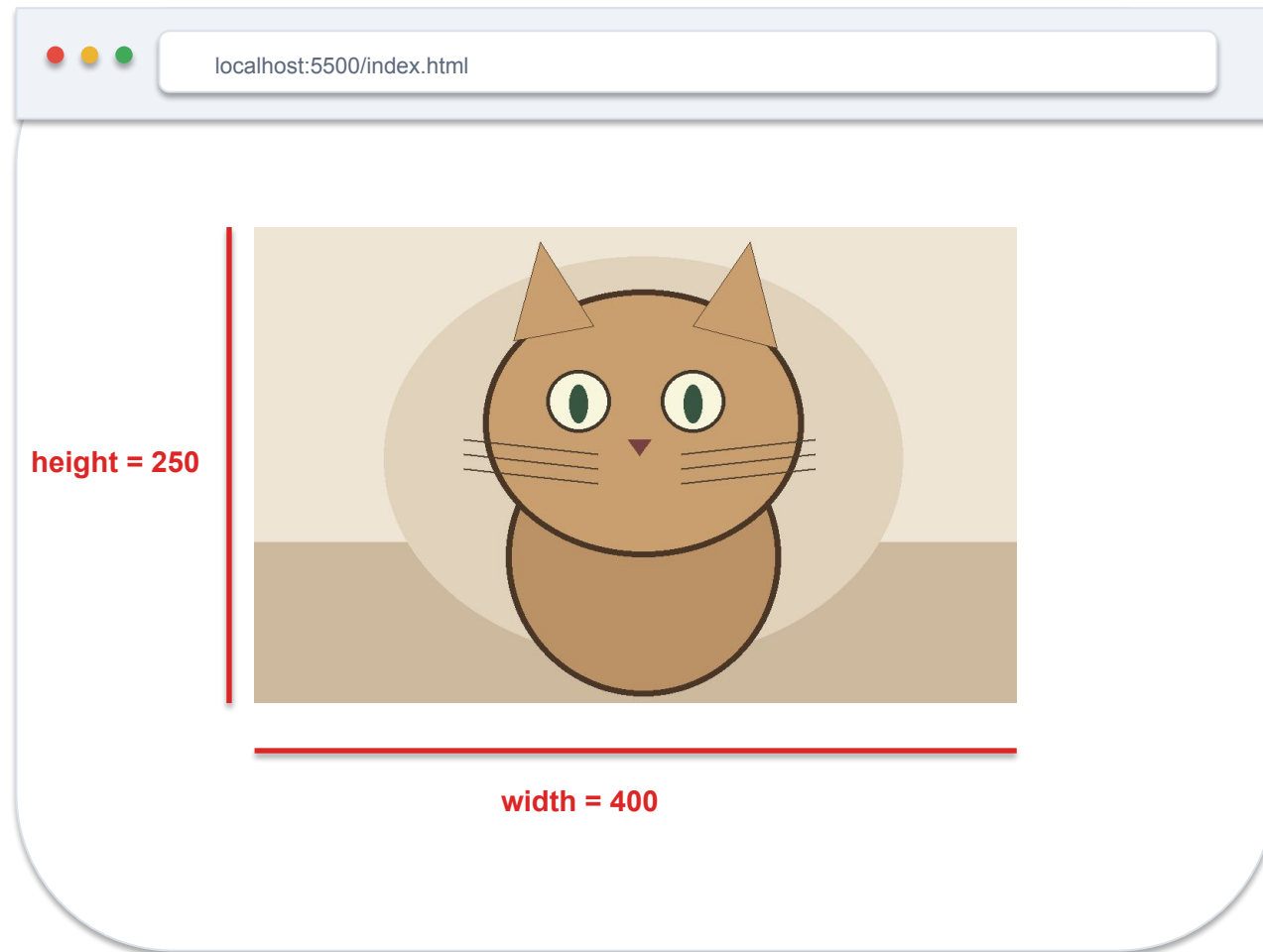
They control the displayed dimensions of the image.

width controls left-to-right size. height controls top-to-bottom size. Numeric HTML values are treated as CSS pixels.

```
index.html  
  

```

Keep the aspect ratio sensible. Random width + height values can stretch the image.



alt Attribute

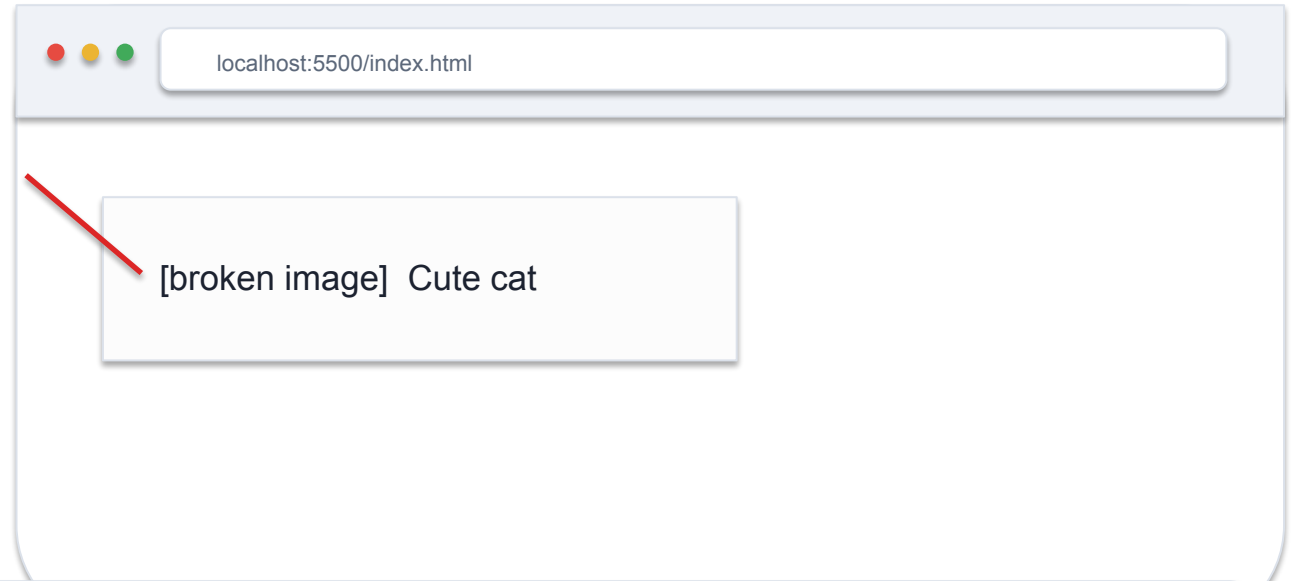
Alternative text describes the image when it cannot be seen.

alt gives a text alternative for meaningful images. It helps screen-reader users and can appear when the image file fails to load.

alt → fallback
description

```
index.html


```



Good alt: "Student writing HTML code on a laptop"

Weak alt: "image" or "photo123"

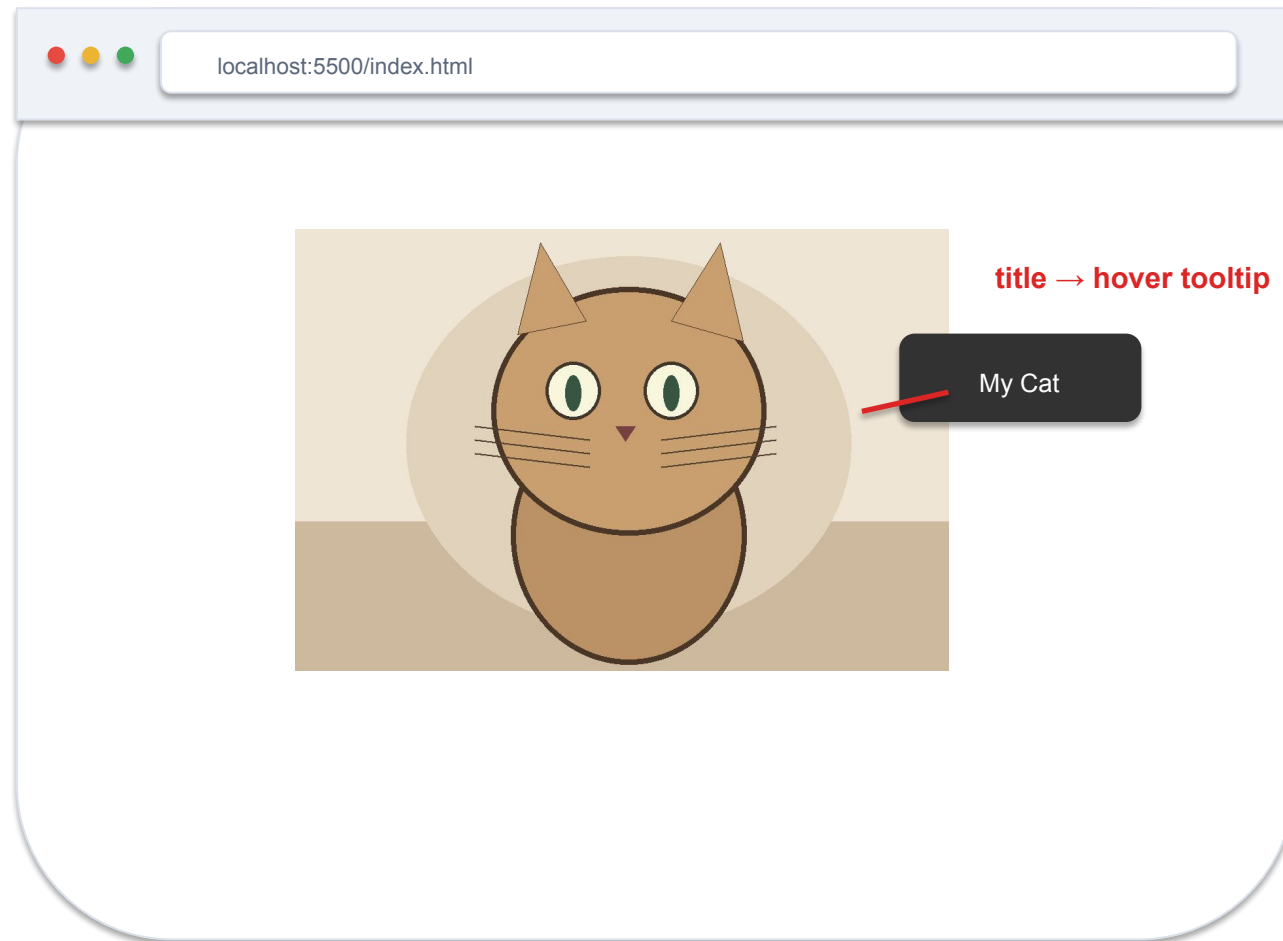
title Attribute

title commonly appears as a tooltip when the pointer stays over the image.

```

```

Do not use title instead of alt. Important information should not depend only on hover.



border, align, hspace and vspace

Older HTML image attributes. Understand them, but use CSS in modern projects.

```
index.html

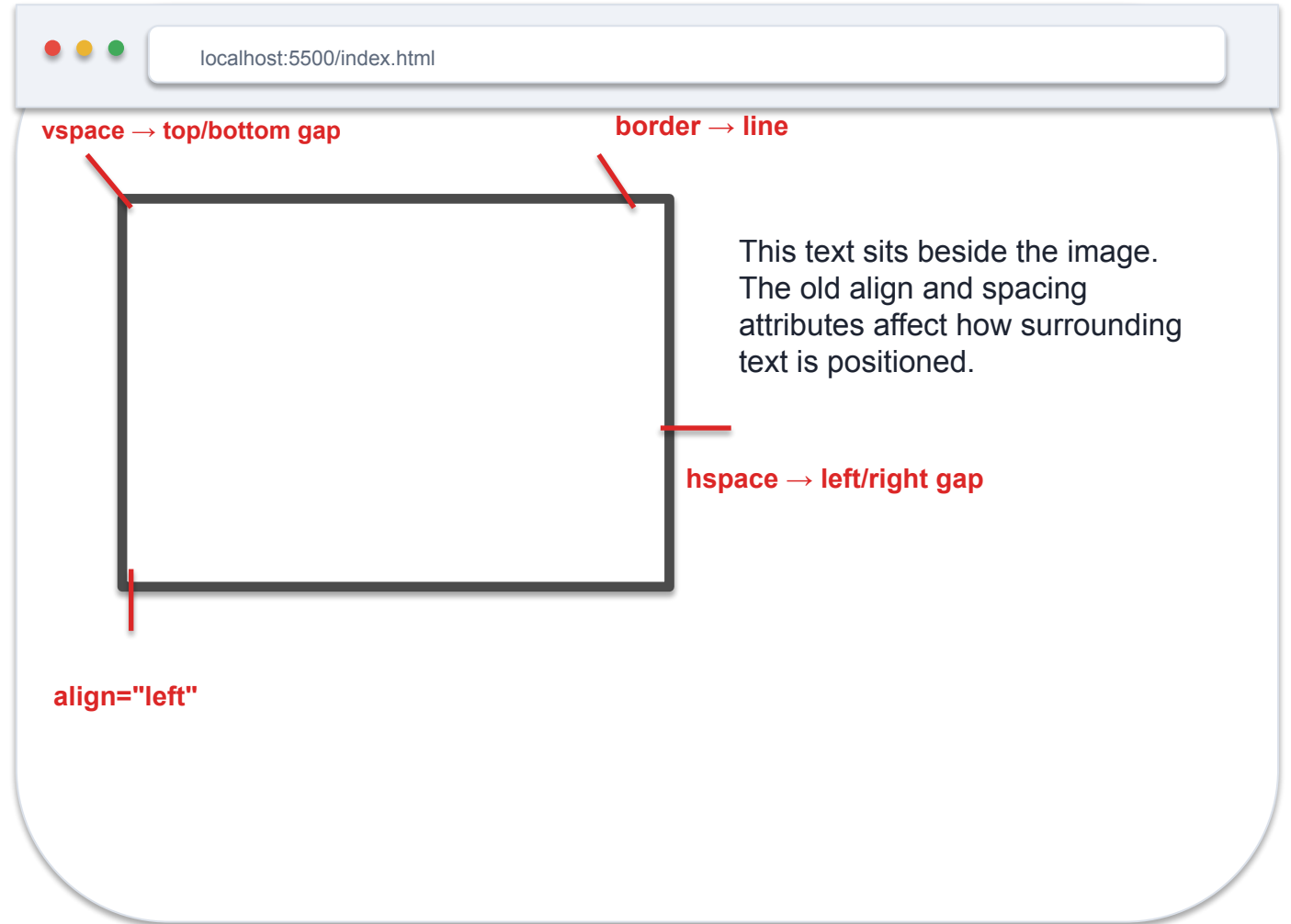

```

Modern replacements

border → CSS border

align → Flexbox/Grid/float where appropriate

hspace / vspace → CSS margin



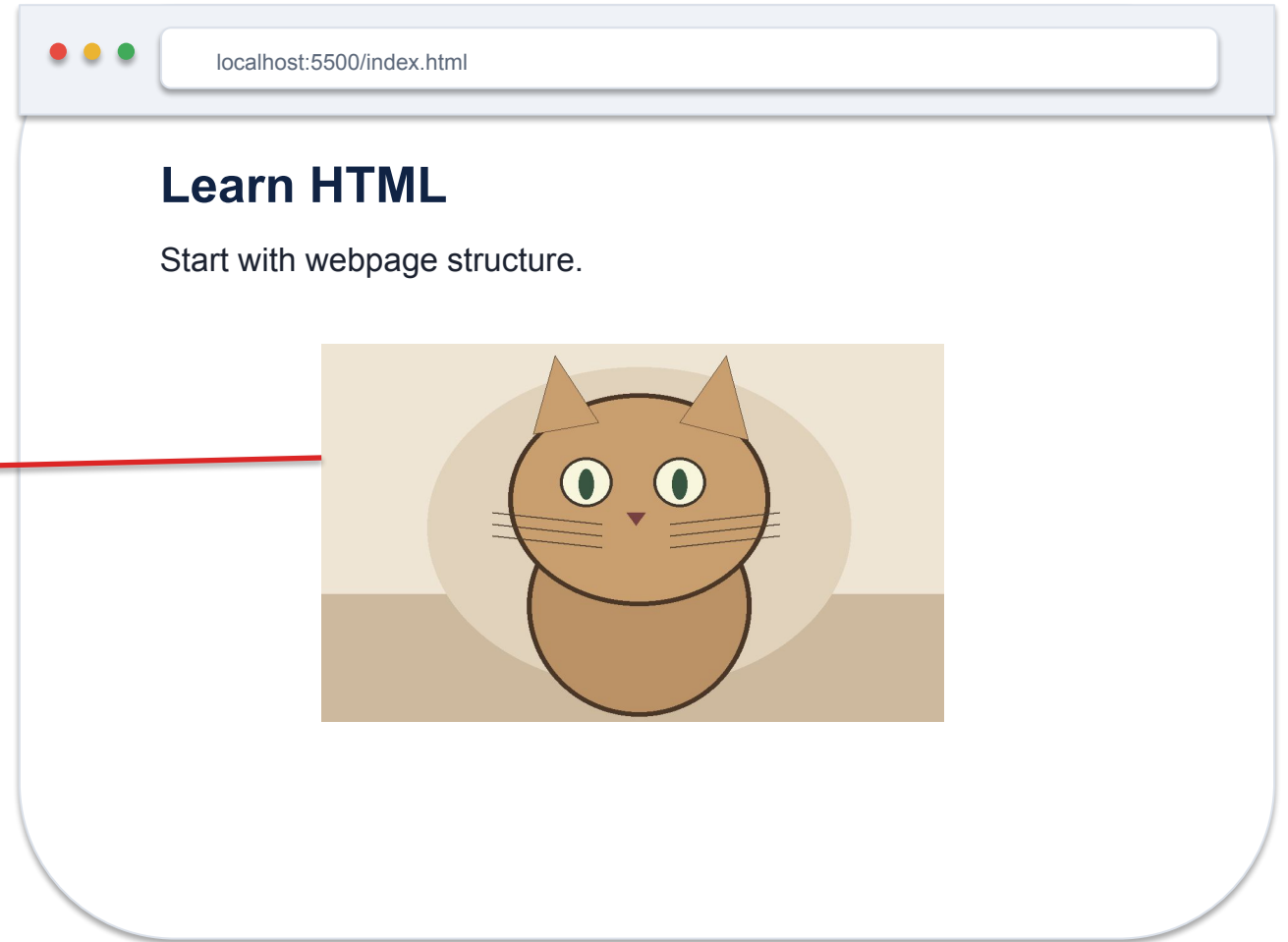
Images Inside the <main> Section

<main> contains the primary content unique to the current page.

```
<main>
  <h1>Learn HTML</h1>
  <p>Start with webpage structure.</p>
  
</main>
```

image is part of main
content

Putting an image inside <main> does not change how works. <main> gives semantic meaning to the page structure.



crossorigin Attribute

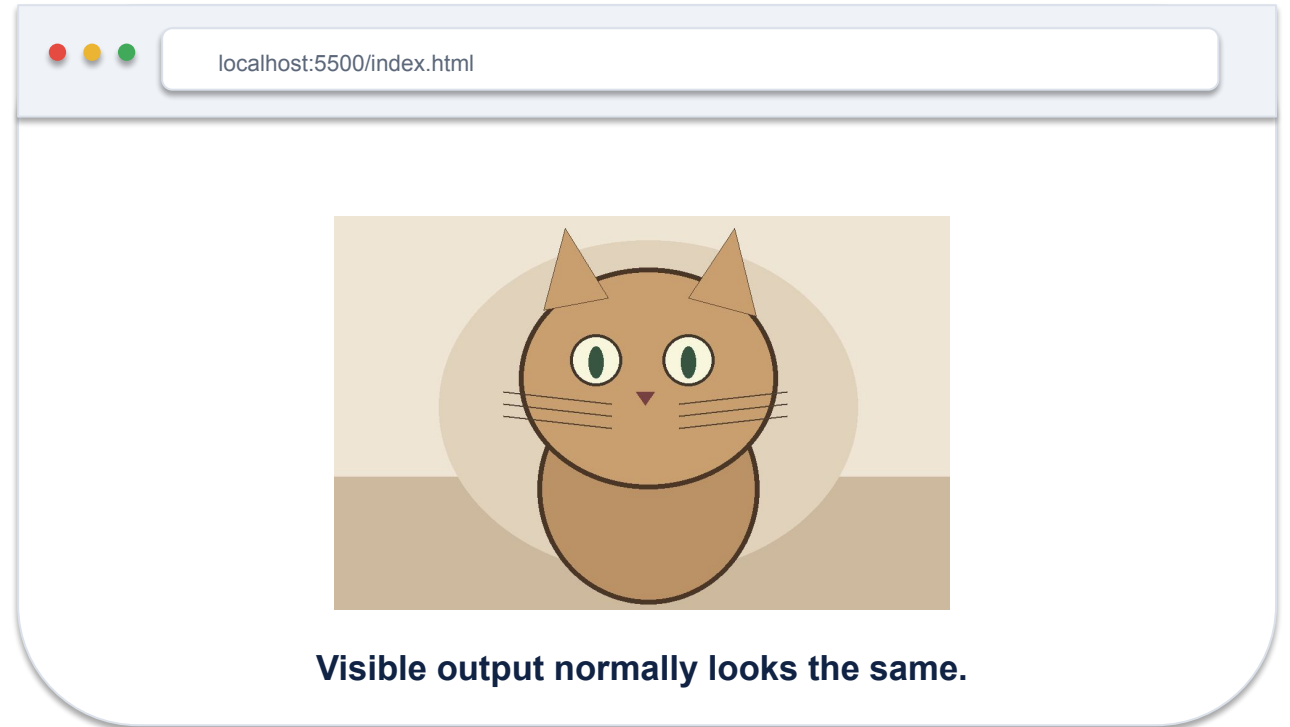
Controls how a cross-origin image request uses CORS.

Use this when an image comes from another origin and the browser needs a CORS-enabled request, especially for cases such as using that image with canvas.

```
index.html


```

- anonymous** CORS request without cross-origin credentials.
- use-credentials** CORS request with credentials when applicable.



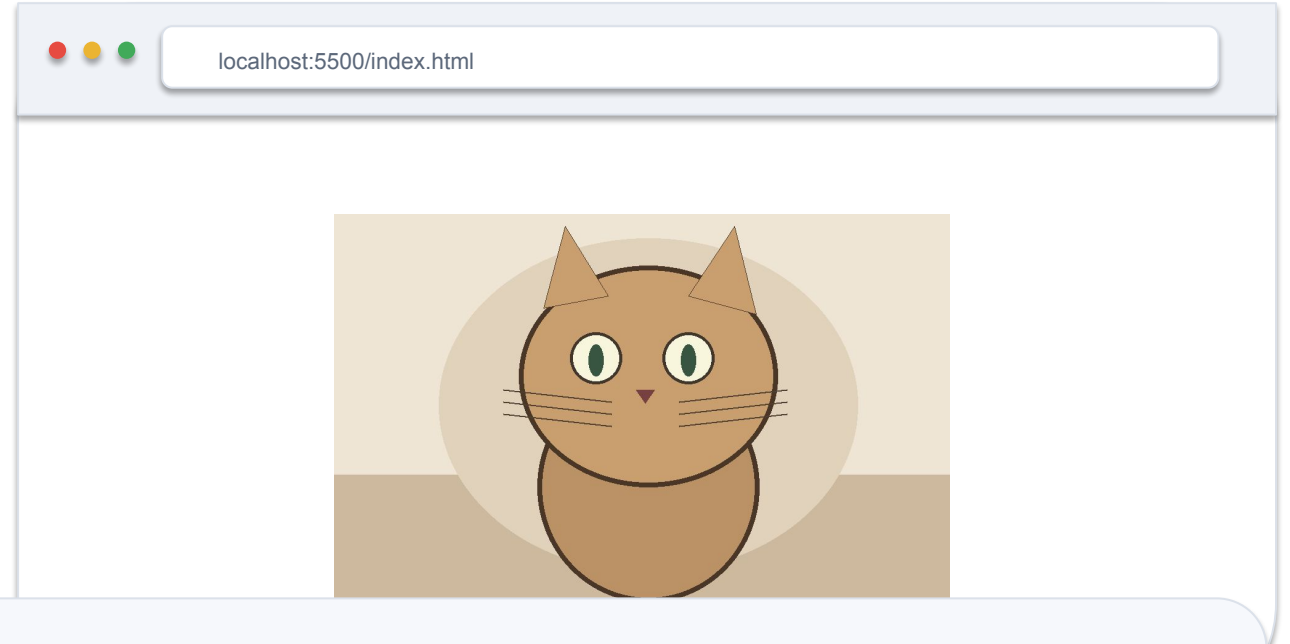
decoding Attribute

A hint about how the browser should decode downloaded image data into pixels.

Flow: download the image file → decode its data → display pixels. decoding changes browser processing, not the visual design of the image.

```
index.html


```



async

Decoding may happen asynchronously.

sync

Hint to decode synchronously.

auto

Browser decides. Default behavior.

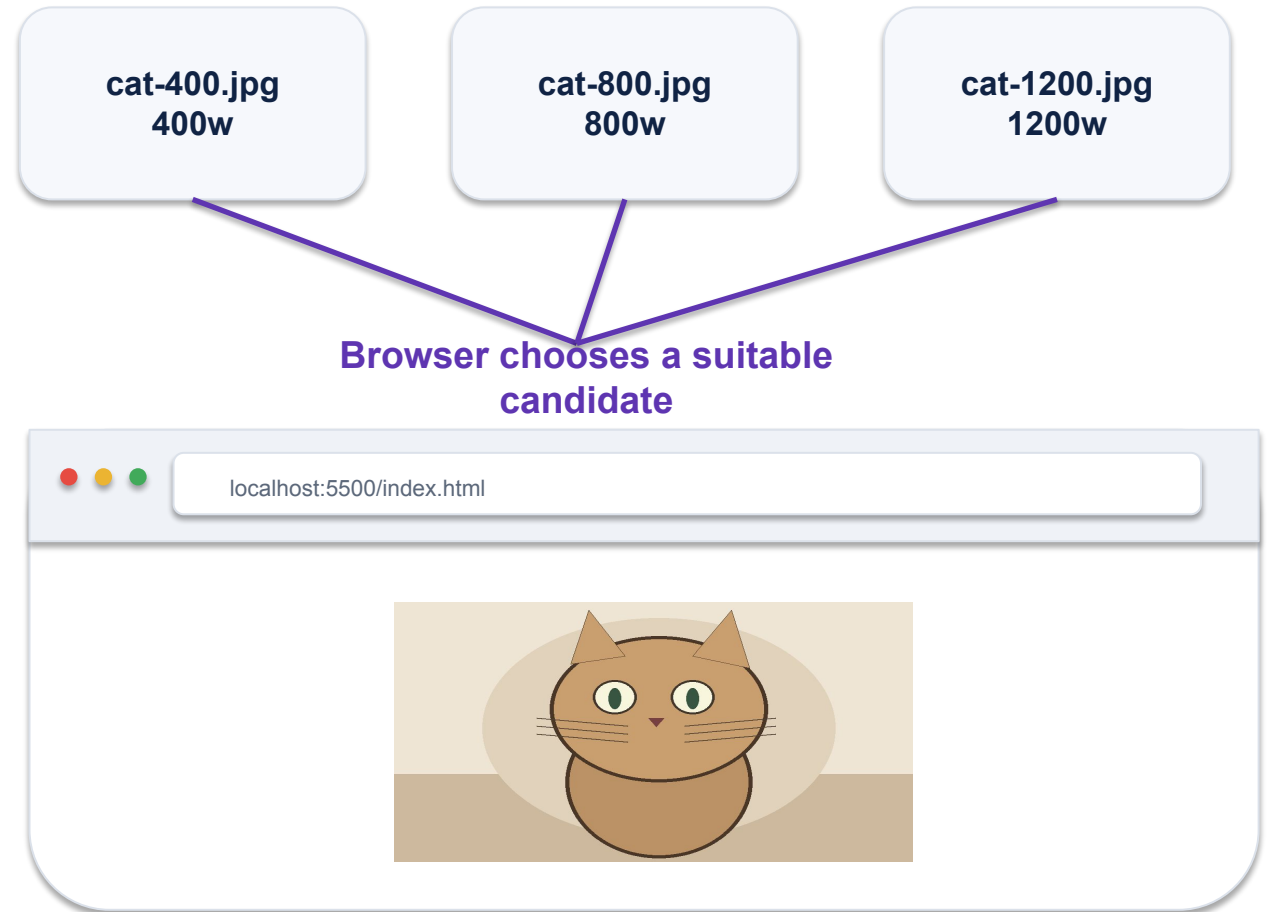
srcset Attribute

Provide multiple image candidates; the browser chooses a suitable file.

```
index.html


```

Selection can depend on the image's display size, viewport and device pixel density. It is not simply "mobile = 400, desktop = 1200".

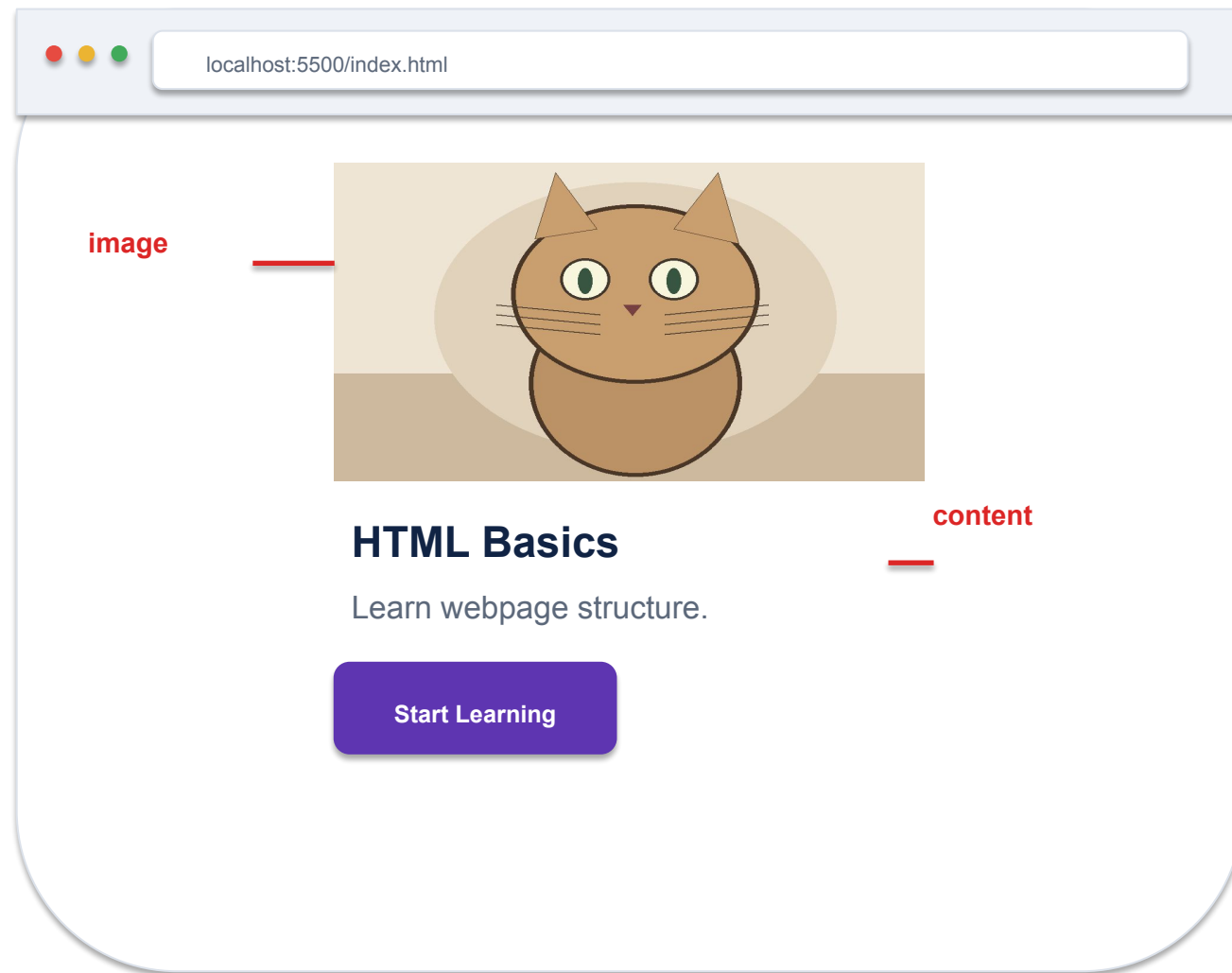


Card with an Image

A card is a UI pattern built from normal HTML elements.

... index.html

```
<article class="card">
  
  <h2>HTML Basics</h2>
  <p>Learn webpage structure.</p>
  <button>Start Learning</button>
</article>
```



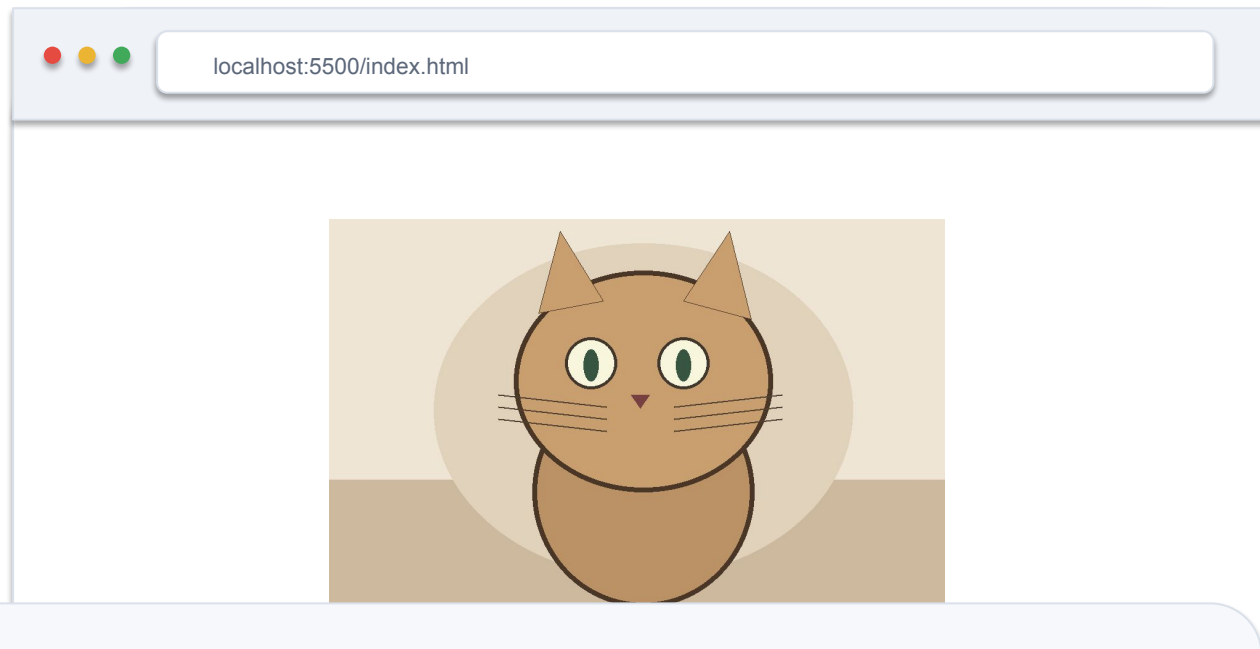
importance → Modern fetchpriority

importance was experimental. For current HTML, teach fetchpriority.

fetchpriority gives the browser a hint about the relative priority of fetching an image. It does not change the image's appearance.

```
index.html  
  

```



high

Relatively important resource.

low

Lower relative fetch priority.

auto

Let the browser decide normally.

Amazon-style Home Page Design

PW IOI

Use images as part of product/category cards. This is a simplified HTML-first layout.

... index.html

```
<main>
  <section>
    <article>
      <h2>Gaming Accessories</h2>
      
      <a href="#">See more</a>
    </article>
  </section>
</main>
```



localhost:5500/index.html

Gaming Accessories



[See more](#)

Shop for Home



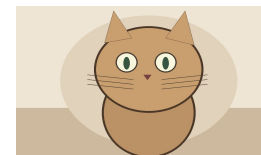
[See more](#)

Top Deals



[See more](#)

New Arrivals



[See more](#)

Concept: repeated cards + headings + images + links

Fluid Image

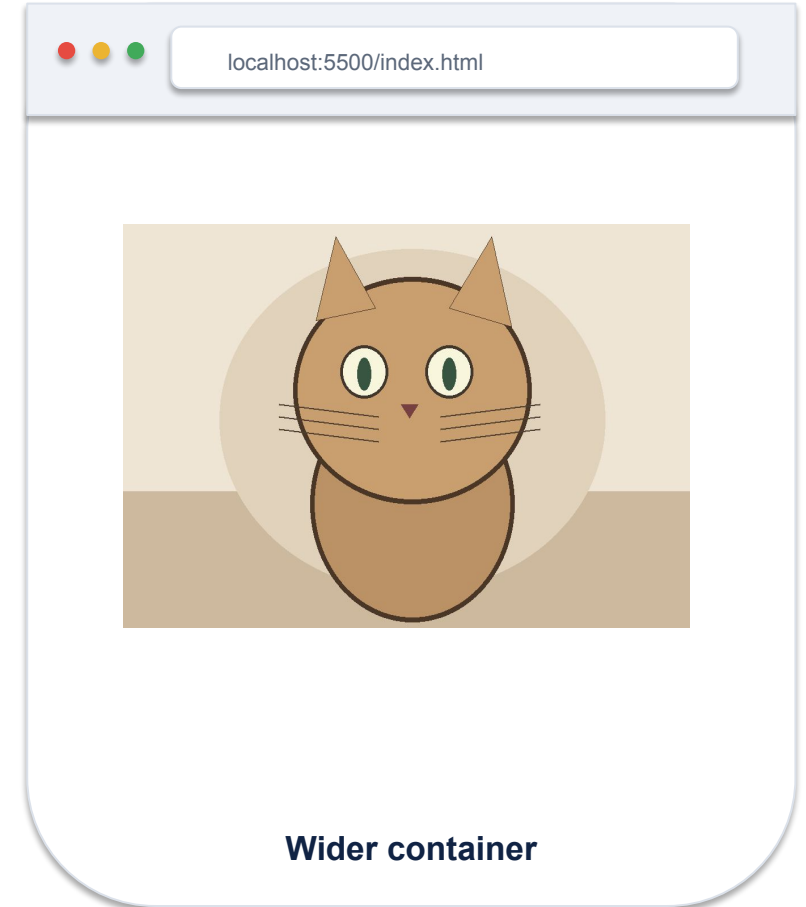
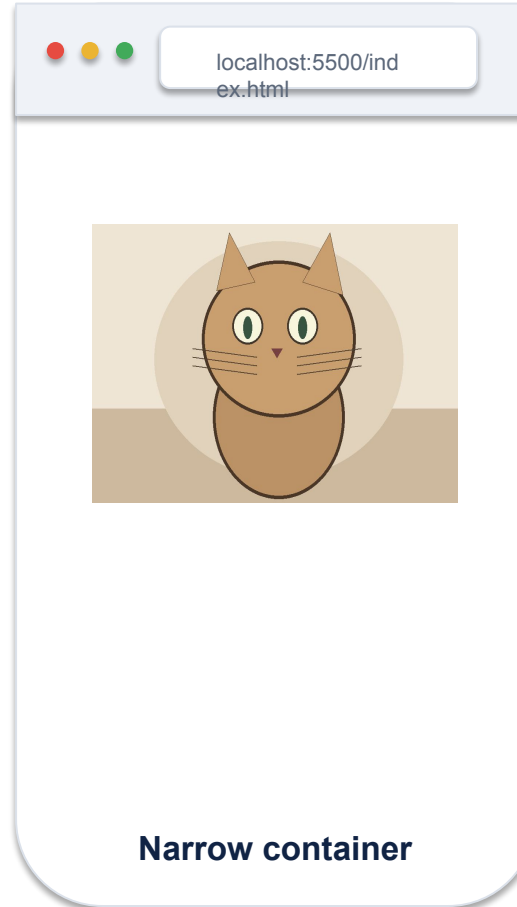
A fluid image scales down with its container while keeping its aspect ratio.

```
index.html



<style>
.fluid-image {
  max-width: 100%;
  height: auto;
}
</style>
```

max-width: 100% → image never becomes wider than its container
height: auto → height adjusts automatically so the aspect ratio is preserved



Card with a Background Image

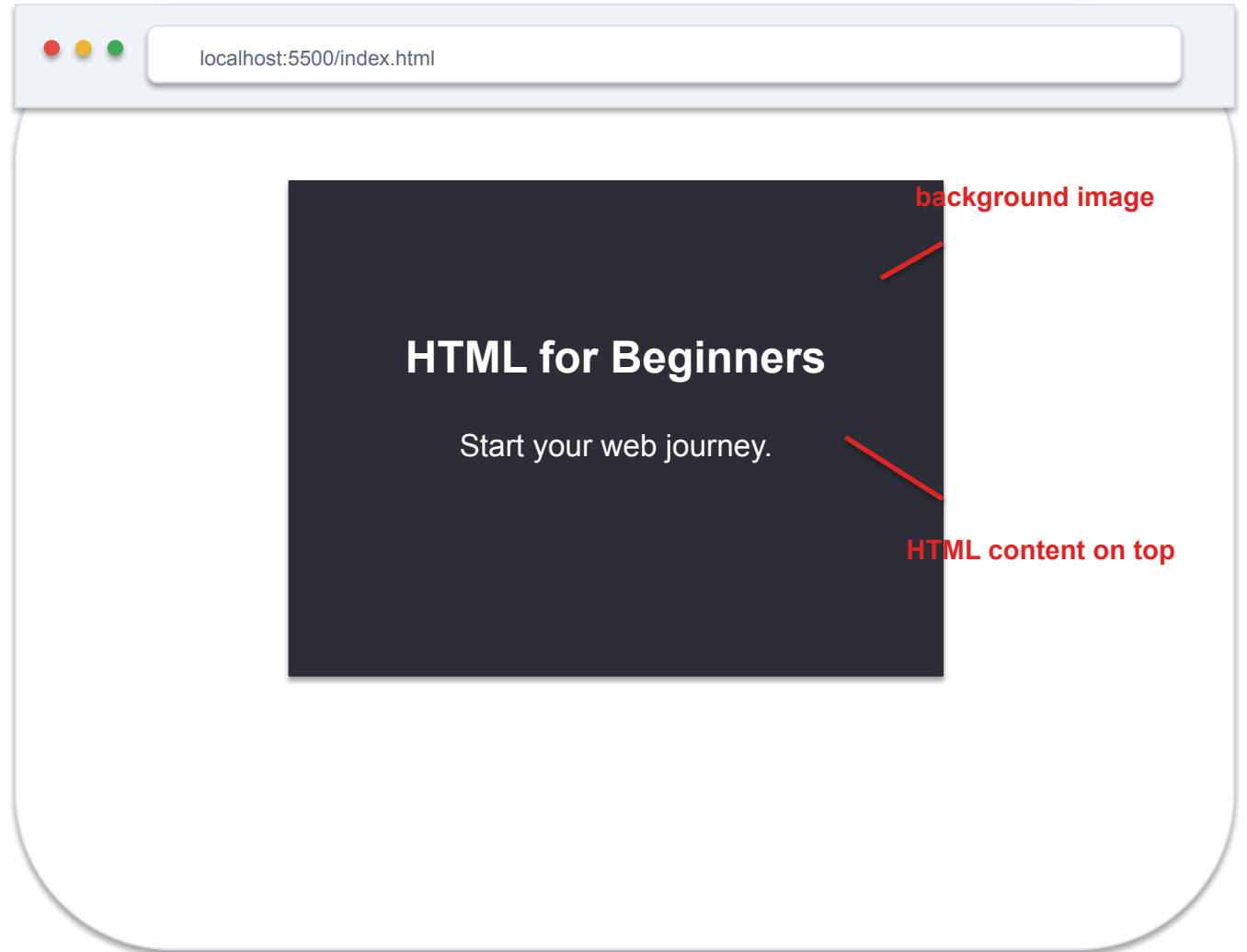
The image is decorative and sits behind the HTML content.

... index.html

```
<div class="course-card">
  <h2>HTML for Beginners</h2>
  <p>Start your web journey.</p>
</div>

<style>
.course-card {
  background-image: url("cat.jpg");
}
</style>
```

Use `<img>` when the image itself is meaningful content. Use `background-image` when the visual is mainly decorative.



Lecture Recap

HTML-first understanding before moving deeper into CSS.

- `<img>` embeds an image; `src` tells the browser where the file is.
- `alt` provides a text alternative; width and height control displayed dimensions.
- `title` can provide a hover tooltip.
- `border`, `align`, `hspace` and `vspace` are legacy attributes; modern projects use CSS.
- `crossorigin` changes CORS request behavior; `decoding` changes decoding hints.
- Teach `fetchpriority` instead of the old experimental `importance` attribute.
- `srcset` provides multiple candidates and lets the browser choose a suitable image.
- Fluid images fit their containers; cards can use `<img>` or decorative background images.